UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL
INSTITUTO DE INFORMÁTICA
CURSO DE ENGENHARIA DE COMPUTAÇÃO

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DNS Centralization and Brazilian Digital Sovereignty: An Economic Perspective

Work presented in partial fulfillment
of the requirements for the degree of
Bachelor in Computer Engineering

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Porto Alegre
July 2025

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ABSTRACT

This study examines the centralization of Domain Name System services and its implications for digital sovereignty, with a focus on Brazil. The research explores how economic and operational dependence on a small number of global DNS providers may affect national autonomy and resilience. To address this issue, a methodology was developed that analyzes both corporate and technical data from major companies in Brazil and the European Union, assessing patterns of DNS dependency and provider distribution. An interactive dashboard was designed to visualize these relationships and illustrate the economic exposure of Brazilian organizations. The results suggest a notable concentration of DNS services among a few international providers in Brazil, contrasting with greater diversity observed in the European context. These findings highlight the potential risks that such centralization poses to digital sovereignty and economic stability, emphasizing the importance of understanding digital dependencies in the management of critical national infrastructure.

Keywords: Digital Sovereignty. DNS Centralization. Economic Dependency. Brazil.

Centralização do DNS e a Soberania Digital do Brasil: Uma Perspectiva Econômica

RESUMO

Este estudo examina a centralização dos serviços de Domain Name System e suas implicações para a soberania digital, com foco no Brasil. A pesquisa explora como a dependência econômica e operacional de um pequeno número de provedores globais de DNS pode afetar a autonomia e a resiliência nacionais. Para abordar essa questão, foi desenvolvida uma metodologia que analisa dados corporativos e técnicos das principais empresas do Brasil e da União Europeia, avaliando padrões de dependência e distribuição dos provedores de DNS. Um painel interativo foi criado para visualizar essas relações e ilustrar a exposição econômica das organizações brasileiras. Os resultados sugerem uma concentração notável dos serviços de DNS entre poucos provedores internacionais no Brasil, em contraste com uma maior diversidade observada no contexto europeu. Esses achados ressaltam os potenciais riscos que tal centralização representa para a soberania digital e a estabilidade econômica, destacando a importância de compreender as dependências digitais na gestão de infraestrutura crítica nacional.

Palavras-chave: Soberania Digital, Centralização de DNS, Dependência Econômica, Brasil.

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LIST OF ABBREVIATIONS AND ACRONYMS

AS	Autonomous System
ASN	Autonomous System Number
AWS	Amazon Web Services
BRL	Brazilian Real
CDN	Content Delivery Network
ccTLD	Country Code Top-Level Domain
DNS	Domain Name System
DDoS	Distributed Denial-of-Service
DoS	Denial-of-Service
EU	European Union
FQDN	Fully Qualified Domain Name
GDPR	General Data Protection Regulation
gTLD	Generic Top-Level Domain
HHI	Herfindahl-Hirschman Index
ICANN	Internet Corporation for Assigned Names and Numbers
IP	Internet Protocol
NS	Name Server
RDAP	Registration Data Access Protocol
TLD	Top-Level Domain
US	United States of America
USD	United States Dollar

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1 INTRODUCTION

Digital sovereignty, especially in the context of computer networks, has emerged as a critical theme in academic discourse, public policy development, and national strategic planning (POHLE; THIEL, 2020). As a modern interpretation of the traditional concept of sovereignty, digital sovereignty seeks to address the growing dependence of states, organizations, and individuals on digital infrastructures governed by a small number of powerful global entities (AZEVEDO et al., 2024). This concept encompasses a broad spectrum of technological dimensions, including infrastructure control, hardware manufacturing, intellectual property management, compliance with standards and regulations, cybersecurity, and the regulation of data flows across jurisdictions.

The notion of sovereignty in the digital domain has gained increased relevance due to the constant evolution of the international political landscape and the technological concentration observed in recent decades (FLORIDI, 2020). Among the various aspects of digital sovereignty, the Domain Name System (DNS) stands out as a foundational element for the internet's functioning. Initially designed as a decentralized and redundant infrastructure, DNS resolution has become progressively centralized under the control of a few organizations and service providers. This trend raises significant concerns about national autonomy, economic dependence, and resilience to failures or malicious attacks.

This bachelor's work aims to examine digital sovereignty from an economic perspective, focusing specifically on the centralization of DNS services and their control by a limited number of actors. While prior research has explored metrics and frameworks to measure market share among DNS providers and has conducted national-level analyses of DNS control (AZEVEDO et al., 2024; BATES et al., 2021), there remains a significant gap regarding the quantification of digital sovereignty through the lens of economic and financial consequences, especially within the Brazilian context. In particular, no studies to date have assessed how economic centralization among DNS providers and potential disruptions might financially impact the Brazilian national economy.

To address this gap, the present work analyzes the economic distribution of DNS services among Brazilian companies and estimates the potential financial repercussions of DNS-related incidents for the Brazilian economy. This involves defining the key steps for an economic assessment and developing a visualization tool that maps and illustrates the economic implications of DNS centralization. Through this approach, the work aims to offer a clearer understanding of national dependencies and risks arising from the con-

centration of digital infrastructure.

To ensure a representative and comparative analysis, the study focuses on the top 1,000 Brazilian companies by revenue, as identified in the *Maiores e Melhores* ranking (Exame, 2025). Additionally, the top 1,000 companies by revenue from the European Union (EU) (CompaniesMarketCap.com, 2025) will be analyzed to establish a comparative baseline that reflects different regulatory, economic, and infrastructural contexts. This comparative approach will allow for a deeper understanding of how DNS dependencies and potential vulnerabilities vary across regions with distinct sovereignty and market dynamics.

The relevance of this research is underscored by two primary motivations. First, considering the critical role DNS plays in the operational continuity of Brazilian businesses, this work evaluates the degree of economic centralization in DNS services and the potential consequences for Brazil's digital sovereignty, especially in the context of increasing dependence on a small number of DNS providers. Second, it addresses the financial dependence of Brazilian enterprises on foreign DNS service providers, examining how this reliance can affect national control over essential digital infrastructure. Furthermore, the study explores the structural causes and possible effects of service centralization, as well as their broader implications for Brazil's economic autonomy and resilience.

The research adopts a comprehensive methodology, including the collection and analysis of corporate DNS usage data, mapping of service providers, identification of national versus foreign dependencies, and assessment of the aggregated economic impact of DNS centralization. The development of an interactive visualization dashboard complements the quantitative findings, enabling a clear and intuitive interpretation of how DNS dependencies are distributed among Brazilian companies and sectors.

Therefore, in summary, the main contributions of this work can be described as follows: *(i)* the proposal and implementation of a methodology to assess the financial impact of DNS service centralization in Brazilian companies, *(ii)* collection and analysis of Brazilian companies' data to create a dataset for further economic and technical analysis, and *(iii)* development of a visualization dashboard designed to represent the economic distribution of DNS dependencies and potential failure impacts.

The remainder of this bachelor's work is organized as follows: Chapter 2 presents the necessary theoretical foundations and reviews the key literature, covering topics such as digital sovereignty, DNS conceptualization and its centralization, the Herfindahl-

Hirschman Index (HHI), and prior research in the field. Chapter 3 details the research design, data collection procedures, the development of the consolidated dataset, and the construction of the visualization dashboard, along with a discussion of methodological limitations. Chapter 4 reports and discusses the main findings, including the analysis of the distribution and concentration of DNS infrastructure, sectoral dependence, quantitative evaluation using the HHI, and the insights provided by the dashboard visualizations. Chapter 5 concludes the work by summarizing the principal contributions and outlining possible directions for future research.

2 BACKGROUND AND RELATED WORK

This chapter presents the concepts necessary to understand the work, as well as positioning it in the context of existing works that address digital sovereignty and DNS concentration. First, we present the concept of sovereignty in the context of computer networks. We also explain what DNS is and how its concentration in the hands of a few organizations has been noted. Then, we list the publications that address the topic and present their contributions to the work.

2.1 Digital Sovereignty

The concept of sovereignty in the context of computer networks is typically used to assert some form of collective control over digital content and infrastructures. However, interpretations, subjects, means and definitions of sovereignty can vary significantly, as contextualized by (COUTURE; TOUPIN, 2019). The modern idea of sovereignty, articulated by thinkers like Machiavelli, Bodin, and Hobbes, refers to the supreme authority exercised over a territory or political entity. Until the 20th century, this concept was predominantly applied to control and authority within clearly defined territorial limits. In today's digital age, sovereignty has taken on new importance in public policy formulation and international relations. In the digital realm, the infosphere does not represent a physical territory, data is not finite, and digital assets primarily belong to private companies influenced by market forces and national governments. Therefore, the digital age compels us to reconsider the nature of sovereignty (FLORIDI, 2020).

Various terms related to sovereignty have been analyzed in the digital sphere, including digital sovereignty, cyber sovereignty, internet sovereignty, technological sovereignty, and data sovereignty. (AZEVEDO et al., 2024) adopts the term "digital sovereignty" as it serves as an umbrella term in the current literature, covering multiple aspects.

As presented by (FLORIDI, 2020), in the realm of computer networks, digital sovereignty refers to the control exercised over the physical layers (resources, infrastructure, devices), code (standards, rules, design), and information (content, data) of digital technologies. This control encompasses the ability to influence and restrict the production, design, use, and outcomes of these technologies.

Countries around the world are making significant efforts to regulate data privacy,

enhance cybersecurity, and assert their national sovereignty in the technological sphere. As (AYDN; BENSGHIR, 2019) unlike the traditional conception of sovereignty, digital sovereignty lacks a consolidated international treaty and is approached differently by each nation based on its political and economic priorities.

The urgency of asserting digital sovereignty increased following Edward Snowden's revelations regarding extensive espionage and surveillance conducted by the United States. In the wake of these revelations, political leaders have utilized the concept of digital sovereignty to advocate for reducing technological dependence on foreign nations. A notable case is Brazil: after the Snowden allegations, then-President Dilma Rousseff proposed a plan to diminish the influence of the United States (US) and its tech giants on the Brazilian internet. Similarly, Canada has invested in improving its digital infrastructure to reduce the traffic of data passing through US networks (COUTURE; TOUPIN, 2019). The growing rivalry between the US and China, along with the COVID-19 pandemic, has further accelerated global efforts to reduce technological and economic dependencies.

The EU has been actively working to safeguard its digital sovereignty and to achieve strategic technological autonomy (BARRINHA; CHRISTOU, 2022). It addresses digital sovereignty through both technical and regulatory measures. On the technical side, this includes ensuring data confidentiality and integrity via standards-compliant hardware, the adoption of open-source software, and localized data processing and storage through the European Federated Cloud Initiative. On the regulatory side, it employs the General Data Protection Regulation (GDPR) (TIPPER; KRISHNAMURTHY, 2022).

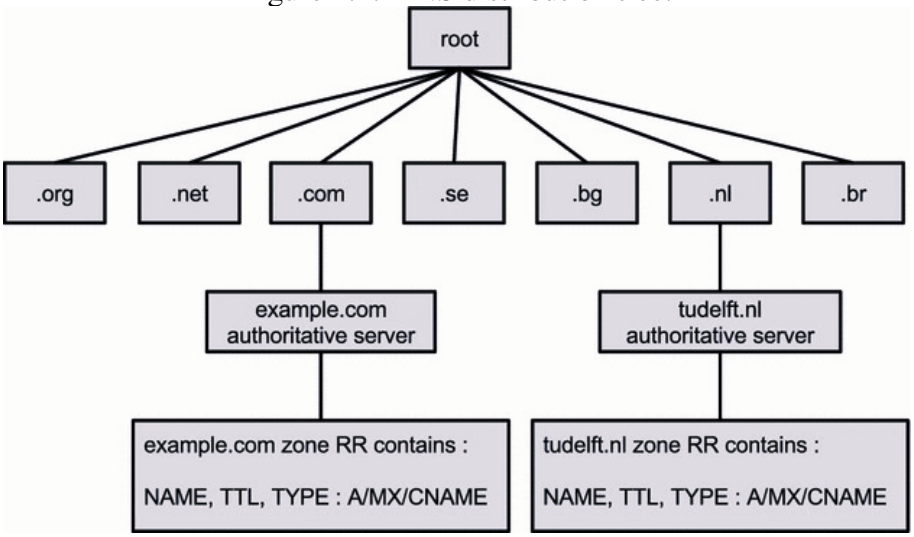
2.2 The Domain Name System

Virtually all modern Internet applications depend on the DNS to function properly. Its main function, translating names into IP addresses, is essential, especially for users, who are accustomed to recognizing "locations" on the internet through friendly words, titles, and abbreviations, rather than numeric IP addresses (DECCIO et al., 2009).

Although the concept of name resolution is relatively straightforward, the system that implements it is highly complex and has broad impacts. Often, the name resolution of a domain depends on servers that are outside the control of the domain owner and are managed by third parties (DECCIO et al., 2009). The large scale of the database and the frequency of its updates make it essential to maintain this system in a distributed manner (MOCKAPETRIS, 1987).

According to (TANENBAUM; FEAMSTER; WETHERALL, 2021), the management of the top of the Internet's naming hierarchy is carried out by the Internet Corporation for Assigned Names and Numbers (ICANN). The conceptual structure of the Internet is organized into top-level domains (TLDs), each housing several hosts. These TLDs are subdivided into subdomains, which can be continuously divided, forming a tree-like naming hierarchy, whose leaves represent domains that do not have subdomains, as seen in Figure 2.1.

Figure 2.1: DNS distribution tree.



Source: (ADRICHEM et al., 2015)

A TLD is a zone located immediately below the root of the DNS hierarchy, exemplified by extensions such as "`.com`" or "`.nl`". TLDs are often classified into subgroups, such as Country Code TLDs (ccTLDs), which represent specific countries or regions (*e.g.*, "`.br`" for Brazil), and Generic TLDs (gTLDs), which encompass generic categories such as "`.org`" or "`.edu`" (HOFFMAN; SULLIVAN; FUJIWARA, 2019).

TLDs are operated by companies called registries, designated by ICANN. For example, Verisign is responsible for registering the ".com" domain. At a lower hierarchical level, registrars are responsible for selling domain names directly to end users. The market has several registrars, which compete with each other by offering different prices and services. Popular examples are: Domain.com, GoDaddy and NameCheap (TANENBAUM; FEAMSTER; WETHERALL, 2021). In Brazil, Registro.br is responsible for the registration and maintenance of domain names that use the "`.br`" (Registro.br, 2025).

The DNS namespace is organized hierarchically. Zones represent parts of this namespace, are managed by a single entity, and are delegated to organizations in a descending manner, starting with the root zone. In the resolution process, name servers, the

resolvers, send queries to authoritative servers, which can provide a direct answer or a referral to a delegated zone (DECCIO et al., 2009). Based on data obtained from OpenIntel, between 2017 and 2021, (ZEMBRUZKI; JACOBS; GRANVILLE, 2022) quantified the numbers of TDLs, domains, NSs and ASs, for both IPv4 and IPv6. Focusing on the results for the end of the period, in 2021 the dataset comprised measurements for 19 TLDs, collectively encompassing 181 million domains. These domains were resolved by approximately 2.8 million distinct IPv4 name servers and around 365,000 IPv6 name servers. In terms of infrastructure, these NSs were distributed across roughly 26,000 autonomous systems (AS) for IPv4 and 4,500 ASs for IPv6. These results for 2021 illustrate the vast scale and complexity of the DNS ecosystem, as well as its continued growth and diversification in both addressing protocols.

As presented by (AZEVEDO et al., 2024), an AS is defined as a network of interconnected computing devices operating under a unified routing policy. Typically managed by a single entity such as an ISP or a technology organization, an AS is identified by an Autonomous System Number (ASN). This structure makes it possible to associate the IP address of any Name Server (NS) with a specific AS, and consequently, with its respective operator and geographical region.

2.3 DNS Centralization

According to (MOURA et al., 2020), in recent years, there has been a growing concern about the centralization of the Internet, which consists of the concentration of services, infrastructure, and users in a few companies. (WANG; MACMILLAN B. SCHAFFNER; CHETTY, 2021) when analyzing the 10 thousand most accessed sites on the Tranco list, concludes that the companies Cloudflare and Amazon alone concentrate more than 30% of DNS domains.

When analyzing data from the (OpenINTEL, 2015) project, and adding it to the prefix-to-AS data and AS-to-organization from CAIDA (CAIDA, 2013; CAIDA, 2014), (ZEMBRUZKI; JACOBS; GRANVILLE, 2022) obtained an overview of the evolution of DNS centralization over the period from 2017 to 2021. The study concludes that in 2021 the concentration of domains using IPv4 from the top 5 DNS ASs reached the 40% mark, with 80% of the domains concentrated in the top 100 ASs of DNS providers.

Such concentration generates vulnerability, as presented by (MOURA et al., 2020), as it creates points of dependence susceptible to attacks, such as the Denial-of-

Service (DoS) attacks against DNS service providers that occurred in 2016, harming Dyn (HILTON, 2016; CHACOS, 2016; BATES et al., 2021), and in 2019, harming AWS (WILLIAMS, 2019). In both cases, important websites and services experienced outages.

2.4 Herfindahl-Hirschman Index

Concentration indexes play a fundamental role in analyzing the structure of markets, offering a quantitative lens through which the distribution of shares or market power among participants can be assessed (SLIME; HAMMAMI, 2016). At their core, these indices serve to measure the degree of inequality, dominance, or dispersion within a given set, whether it refers to an economic sector, a financial portfolio, or as in this case, digital infrastructure systems.

According to (BECKER; DULLMANN; PISAREK, 2004), an effective concentration index must adhere to six essential mathematical and economic properties. The first states that when a share is transferred from a smaller participant to a larger one, without changing the total, the index should increase, reflecting higher concentration. The second criterion establishes that concentration is minimized when all shares are equal, representing maximum equality and dispersion. Third, when comparing two groups with the same number of participants, the group in which the largest shares account for a greater portion of the total should exhibit a higher level of concentration. The fourth criterion indicates that when shares or exposures are combined, overall concentration must rise, since fewer participants come to control a larger part of the whole. The fifth states that when all shares are identical, increasing the number of participants should lead to lower concentration, highlighting the effect of diversification. Finally, the sixth criterion ensures that adding an extremely small share should not significantly affect the index, thus preventing irrelevant values from distorting the measurement of concentration.

The index chosen for this study, the Herfindahl-Hirschman Index (HERFINDAHL, 1950; HIRSCHMANN, 1964), is a widely adopted metric for quantifying the degree of market concentration in empirical studies, and meets all six criteria listed above (SLIME; HAMMAMI, 2016). The HHI is calculated by summing the squares of the market shares, expressed as percentages, of all participants operating in the relevant market. Mathematically, the index is defined as:

$$HHI = \sum_{i=1}^{N} s_i^2 \tag{2.1}$$

Where s_i represents the market share of participant i in percentage terms, and N is the total number of participants in the market. The value of the HHI ranges from close to zero to 10,000. An HHI close to zero corresponds to a completely fragmented market with an extremely large number of small participants, each holding an insignificant share. At the opposite extreme, an HHI close to 10,000 characterizes a pure monopoly, where a single firm holds the entire market share.

To illustrate, consider a market composed of four firms with market shares of 40%, 30%, 20%, and 10%. Applying the formula, the HHI is calculated as presented in Equation 2.2.

$$HHI = 40^2 + 30^2 + 20^2 + 10^2 = 1600 + 900 + 400 + 100 = 3000 \tag{2.2}$$

The interpretation of HHI values provides clear guidance for competition analysis. According to thresholds recommended by the US Department of Justice and the US Federal Trade Comission, which are adopted by its Antitrust Division, a market with an HHI below 1,000 is considered unconcentrated, indicating high competition. Values between 1,000 and 1,800 denote a moderately concentrated market, while an HHI equal to or greater than 1.800 signals high concentration and reduced competitiveness (U.S. Department of Justice, 2024; U.S. Department of Justice and the Federal Trade Commission, 2023).

2.5 Literature Review

(BATES et al., 2021) uses the three consecutive Distributed Denial-of-Service (DDoS) attacks against the DNS provider Dyn in 2016 as a framework to analyze the resilience of DNS structures, focusing on the increasing concentration of DNS services and the low utilization of redundancy. As part of the study, the author sampled the top 100 US domains under the "`.com`," "`.net`" and "`.org`" TLDs, according to the Alexa Top Sites list. The research spanned the period from 2011 to 2018, with data collected monthly. The study utilized the CompleteDNS API to gather historical information on

the NS associated with each domain. To evaluate market share concentration, the study employed the HHI, calculated by squaring the market share of each firm competing in the market and then summing the resulting values.

As a result, the study identified an increase in the HHI over the analyzed period. One inference made by the author, based on the study, is that the growing outsourcing of services to external providers is one of the factors contributing to the concentration of services among a few companies. During the period of the attacks, Dyn held approximately 10% of the market share.

In another study, (ZEMBRUZKI; JACOBS; GRANVILLE, 2022) investigated the global concentration of DNS services using data from the OpenIntel project, which stores information on over 60% of the global DNS. OpenIntel collects daily data from complete zone files of popular gTLDs like "`.com`", "`.net`", and "`.org`", as well as approximately 16 ccTLDs, including Russia ("`.ru`"), Germany ("`.de`"), and China ("`.cn`"). One of the objectives of the research was to analyze a wide range of data, further than in previous works. Therefore, the dataset analyzed included 147 to 181 million domains distributed across 11 to 19 TLDs, resolved by approximately 2.4 to 2.8 million IPv4 name servers and approximately 365,000 IPv6 servers, served by approximately 26,000 ASs using IPv4 and 4,500 ASs using IPv6. The study further enriched the analysis by mapping NS IP addresses to ASs using the CAIDA database, linking ASs to organizations and countries for a detailed view of the centralization and geographic dependencies of DNS services.

The research analyzed the concentration of the DNS market over five years (2017-2021), segmenting domains by year and TLDs, focusing on market share distribution and the geographic location of dominant companies, particularly US-based providers. In 2017, the five largest DNS ASs resolved 30% of domains using IPv4, increasing to 40% by 2021. Furthermore, the top 100 ASs concentrated approximately 80% of the 1.36 million domains analyzed. These findings reveal a growing reliance on a few DNS service providers, raising concerns about digital resilience and sovereignty due to the risk of systemic failures and over-dependence on limited infrastructure.

Another relevant contribution by (ZEMBRUZKI, 2020) is the development of an active measurement solution to assess DNS centralization, using the `dnstracker` tool to map routes to authoritative servers for the Alexa Top 1 Million domains. Their findings highlight significant infrastructure sharing, with up to 12,000 authoritative servers relying on the same provider, creating single points of failure and exposing many domains to collateral damage in case of attacks. The study also reveals that over 70% of domains do not

adopt redundancy practices, making them particularly vulnerable. Although no increasing trend in centralization was observed during the five-month measurement period, the tool and methodology developed in this research represent a valuable advance, providing network operators and researchers with accessible means to assess and visualize DNS centralization.

To assess national dependence on foreign DNS providers, (AZEVEDO et al., 2024) analyzed how ccTLDs are resolved in two major economic blocs: the BRICS nations, Brazil, Russia, India, China, and South Africa, and the EU.

The authors used a measurement approach, previously proposed by (BOEIRA et al., 2024), to map popular Internet domains to the NS that resolve them, identifying the AS responsible and the organizations behind these services. The process is based on DNS record data and AS information, such as the most accessed domains in the Tranco List, and the CAIDA database, which provides details about ASs and their associated organizations. Data collection starts with obtaining an updated list of ASs and popular domains. From there, DNS records are retrieved for each domain and mapped to the IP addresses of the name servers. This allows the association of each name server with an AS and, subsequently, with the organization managing it, identifying its geographical location and the regulations it is subject to. The NS resolution flow generated by the approach illustrates the path from the domain to the organization managing the infrastructure. This is crucial for understanding the potential risks of centralization and digital sovereignty, as NS and AS operators can influence DNS resolution operation and security.

Taken together, these studies highlight a clear and consistent trend toward the concentration of DNS infrastructure across different contexts and scales. (BATES et al., 2021) demonstrates how the lack of redundancy and increasing outsourcing to third-party providers have resulted in a heightened vulnerability among major domains, as evidenced by the rise in market concentration metrics and the impact of DDoS attacks. (ZEMBRUZKI; JACOBS; GRANVILLE, 2022), using a much broader dataset, reinforces the concern of global DNS centralization, revealing the predominance of a few large service providers and the potential for systemic risks. Their active measurement work, especially through the `dnstracker` tool, also exposes the significant infrastructure overlap and the lack of redundancy strategies, further amplifying the risks associated with centralization. (AZEVEDO et al., 2024) expand the discussion to the geopolitical domain, illustrating how national and economic blocs depend on foreign DNS operators, raising issues of digital sovereignty and regulatory exposure. Collectively, these contributions

underscore that centralization in DNS is not only a technical and operational risk but also a question of resilience, governance, and autonomy for both organizations and countries. This consolidated body of work sets a comprehensive foundation for the present analysis, which aims to further explore the evolving landscape of DNS centralization and its broader implications in digital governance, focusing in the economic perspective.

3 METHODOLOGY

In this chapter, we present the methodological procedures adopted to investigate the DNS centralization in its economic perspective. The methodology was structured to ensure rigor, transparency, and reproducibility, integrating sequential stages of data extraction, technical enrichment, and analytical treatment.

3.1 Research Design and Analytical Strategy

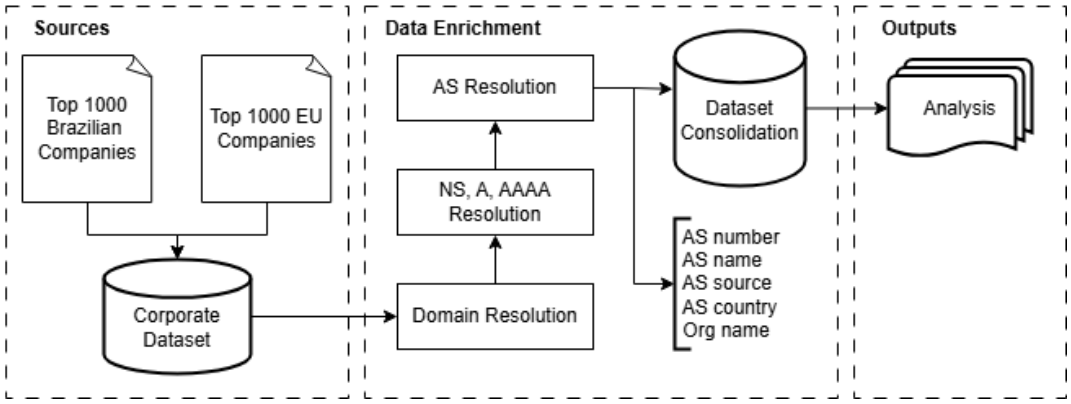
This study adopts an empirical-analytical approach to examine critical issues related to digital sovereignty and the centralization of DNS resolution infrastructure. The analysis is centered on a dataset constructed from the 1,000 largest companies in Brazil, ranked by annual revenue and sourced from the *Exame Melhores e Maiores* ranking(Exame, 2025), and a corresponding dataset of the 1,000 largest companies in the EU, obtained from Companies Market Cap(CompaniesMarketCap.com, 2025), also based on revenue.

The methodological premise is that, due to their economic relevance and digital footprint, these leading companies provide a privileged perspective on the structure of network infrastructure at both national and regional levels. The focus on revenue as the main selection criterion is justified by its established use as a proxy for market share in international best practices, including guidelines from the US Department of Justice and Department of Trade for antitrust and merger analysis (U.S. Department of Justice, 2024; U.S. Department of Justice and the Federal Trade Commission, 2023). Revenue is thus recognized as a robust metric for mapping corporate influence and sectoral dominance within the digital ecosystem.

3.2 Pipeline Overview and Data Collection Process

The methodological pipeline, as seen in Figure 3.1, is organized into three main phases: `(i)` extraction of company information and financial metadata, and identification of the official domains for each company; `(ii)` technical enrichment with DNS and network infrastructure data; and `(iii)` the construction of a structured database for subsequent analysis. Each phase is detailed in the following subsections.

Figure 3.1: Data pipeline.



Source: The Author, 2025

3.2.1 Data Sources and Initial Collection

The first phase centers on selecting and collecting data on the top companies in each region. For Brazil, the corporate list was obtained from the *Exame Melhores e Maiores* annual ranking, which provides revenue and sector information for each company. For the EU, the selection was based on the ranking of companies by annual revenue, extracted from the Companies Market Cap platform. Because the European dataset did not provide explicit sector classifications, a transposition of the Brazilian sector mapping was implemented. To achieve this, we used the DeepSeek R1 artificial intelligence model (DeepSeek, 2024), applying advanced search and reasoning capabilities to infer sector categories for European companies based on their profile and business description.

Once the company lists and sectoral mappings were finalized, the next step was to identify the official domain of each company. This was accomplished through an automated search routine using the `duckduckgo-search` search package in Python. The automated process generated custom queries for each company, extracting the fully qualified domain name (FQDN) using regular expressions. This ensured that only active, publicly accessible domains were retained in the working dataset.

3.2.2 DNS Resolution Data

The technical enrichment of the dataset involved a systematic process of DNS resolution and network attribution, implemented using Python and widely adopted open-source libraries (PINTO et al., 2024). In this stage, the primary goal was to map the entire

DNS chain for each domain identified in the previous phase. As shown in Listing 3.1, the process began with the use of the method `get_ns_info`, which received the domain as input and, through the `dns.resolver` library, extracted all authoritative NS associated with that domain. Since many domains can have multiple NSs, the routine was designed to capture the complete set, reflecting the full scope of the DNS infrastructure supporting each company.

Listing 3.1 – Function to obtain NS and ASN information for a domain.

```
def get_ns_info(domain):
    ns_data = []
    try:
        ns_answers = dns.resolver.resolve(domain, 'NS')
        for ns in ns_answers:
            ns_name = str(ns.target).rstrip('.')
            # Try to get IPv4 and IPv6
            ns_ip = get_ip(ns_name, 'A')
            ns_ipv6 = get_ip(ns_name, 'AAAA')
            ip_to_use = ns_ip or ns_ipv6
            if ip_to_use:
                try:
                    whois_obj = IPWhois(ip_to_use)
                    result = whois_obj.lookup_rdap()
                    asn = result.get("asn")
                    as_name = ... # Parse ASN description
                    as_country = result.get("asn_country_code")
                    org_name = extract_org_name(result)
                except:
                    asn = as_name = as_country = org_name = None
            else:
                asn = as_name = as_country = org_name = None
            ns_data.append({
                "NS": ns_name,
                "NS-A": ns_ip,
                "NS-AAAA": ns_ipv6,
                "ASN": asn,
                ...
            })
        return ns_data
    except Exception as e:
        # Handle exceptions and try trimmed domain
        ...
```

For each NS obtained, the method initiated a loop that queried for both `A` and `AAAA` records using the same `dns.resolver` library. The `A` records correspond to IPv4 addresses, while the `AAAA` records refer to IPv6 addresses. Collecting both types of records was essential to accurately represent the adoption of traditional and modern network protocols. This comprehensive mapping of all reachable IP addresses for each NS was a critical step, as it laid the groundwork for subsequent analysis of network ownership and infrastructure concentration.

With the list of IP addresses assembled, the workflow advanced to the enrichment of network attribution data. The `IPWhois` library was used to retrieve the full Registration Data Access Protocol (RDAP) object associated with each address. RDAP is a protocol that provides structured registration information for IP addresses and AS numbers, improving on traditional `IPWhois` by offering hierarchical and machine-readable data (ICANN, 2025). As seen in Listing 3.2, by parsing the RDAP response, it was possible to extract attributes such as the ASN, the name and source of the AS, the country of registration, and, importantly, the organization name linked to the allocation of each IP block.

Listing 3.2 – Compact RDAP/WHOIS result for an IPv4 address.


```
{
  "asn": "15169",
  "asn_description": "GOOGLE, US",
  "asn_registry": "arin",
  "asn_country_code": "US",
  "query": "74.125.225.229",
  "network": {
    "cidr": "74.125.0.0/16",
    "ip_version": "v4",
    "handle": "NET-74-125-0-0-1",
    "name": "GOOGLE",
    "start_address": "74.125.0.0",
    "end_address": "74.125.255.255",
    "type": "DIRECT ALLOCATION",
    "status": ["active"],
    "links": [
      "https://rdap.arin.net/registry/ip/74.125.0.0"
    ]
  },
```

```
  "objects (Org/Registrant only)": [
    {
      "handle": "GOGL",
      "type": "org",
      "name": "Google LLC"
    }
  ]
}
```

To ensure accuracy in associating NS to their operating organizations, special attention was given to the fields describing the registrant and administrative contacts within the RDAP object. In situations where name resolution was ambiguous, particularly due to the use of subdomains or variations in the FQDN, a recursive procedure was implemented. This approach progressively simplified the domain, stripping leftmost labels and reattempting the DNS and RDAP queries to maximize the likelihood of a successful match. This strategy substantially increased the completeness of the dataset.

The entire technical pipeline was applied consistently to both the Brazilian and European datasets, ensuring methodological symmetry and enabling robust comparative analysis. As a result, each domain record was enhanced with detailed technical metadata, including the complete set of authoritative NS, associated IP addresses for both IPv4 and IPv6, and enriched information regarding the underlying AS and responsible organization.

3.2.3 Final Dataset Consolidation

The final stage of the methodological pipeline consisted of the construction of a structured database for analysis, consolidating all information extracted during the data collection, financial mapping, and DNS resolution phases. After harmonizing the variables, two main tables were created: one for Brazil and another for the EU.

The structured dataset developed for this research provides a comprehensive overview of the technical, geographic, and financial attributes associated with each company included in the study. For both Brazil and the EU, the dataset was organized to consolidate the main variables necessary for the subsequent analyses. As shown in Table 3.1, the structure adopted ensures methodological consistency between the regions, while reflecting the specificities of the available data for each context.

It is important to note that the aggregated table was built by merging the financial

Table 3.1: Structure of the Consolidated Dataset

Field	*Type*	*Description*
company_name	Object	Company name
country_name	Object	Country where the company is headquartered
state_name	Object	State of headquarters (Brazil only)
city_name	Object	City of headquarters (Brazil only)
sector_primary	Object	Primary sector of activity
revenue	Float64	Annual revenue
earnings	Float64	Earnings
total_assets	Float64	Total assets
domain	Object	Official domain name
ns	Object	NS
ns_a	Object	IPv4 address (A record) of the NS
ns_aaaa	Object	IPv6 address (AAAA record) of the NS
asn	Object	AS Number
as_name	Object	Name of the AS
as_source	Object	Source registry of the AS
as_country	Object	Country of the AS
org_name	Object	Organization responsible for the AS

Source: The Author, 2025

data and DNS resolution outputs, resulting in multiple entries for each company or, more specifically, each domain. Each record refers to a single NS identified and resolved for a given domain. For Brazil, considering the top 1,000 companies, the final table contained 7,260 entries. Based on this dataset, a consolidation process was carried out to allow for a centralized and unified analysis of all collected information.

Various aggregation operations were performed to construct a final dataset with an expanded structure, enabling more advanced and comparative assessments. The core fields such as `company_name`, `country`, `state`, `city`, and `revenue` were maintained. Then, new atributes were added, as presented in Table 3.2.

Additional attributes provide details about national reliance in DNS resolution. There are also flags related to IPv6 support, such as `has_ipv6_ns`, and a composite flag reflecting dual-stack presence. A `dominant_provider` column was included to define the dominant provider for each company. Since it is not feasible within the limits of this research to map every technical role of each NS, such as distinguishing between servers used for HTTP and those used for email, the identification of the dominant provider was based on the one with the greatest share of DNS responsibility for the company.

This set of consolidated variables also includes fields specific to the dominant

Table 3.2: Additional Fields in the Final Consolidated Dataset

Field	*Type*	*Description*
num_ns	Int64	Number of NS
num_asn	Int64	Number of distinct ASs used by the NSs
asn_list	Object	List of ASs associated with the NSs
num_country	Int64	Number of countries where NSs are hosted
countries_used	Object	List of countries where NSs are hosted
pct_national_ns	Float64	Percentage of NSs hosted in homecountry
only_foreign_ns	Bool	True if all NSs are hosted abroad
has_ipv4_ns	Bool	True if at least one NS has an IPv4 address
has_ipv6_ns	Bool	True if at least one NS has an IPv6 address
dual_stack	Bool	True if at least one NS supports IPv4 and IPv6
dominant_provider	Object	Provider responsible for the majority of NSs
dominant_provider_country	Object	Country of the dominant provider
dominant_provider_asn	Object	ASN of the dominant provider
only_one_provider	Bool	True if only one provider is used for all NSs
distinct_asns	Int64	Number of distinct ASNs among NSs
distinct_countries	Int64	Number of distinct countries among NSs
num_ns_in_brazil	Int64	Number of NSs located in Brazil
num_ns_foreign	Int64	Number of NSs located abroad
infra_type	Object	Type of DNS infrastructure used

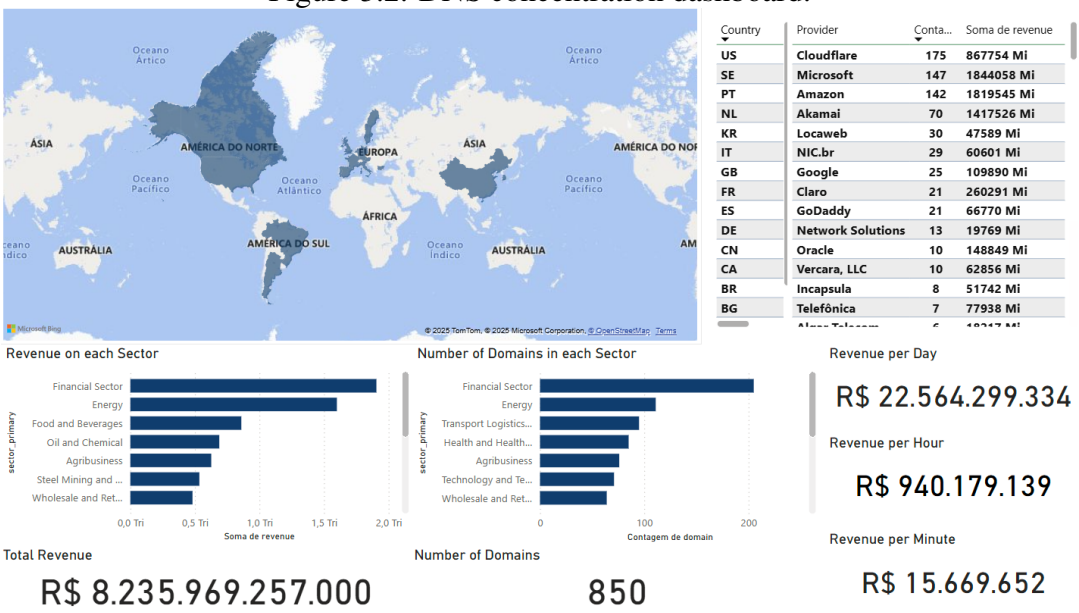
Source: The Author, 2025

provider, such as `dominant_provider_country`, `dominant_provider_asn`, and the Boolean field `only_one_provider`, which indicates if a company is fully dependent on a single DNS provider. With this enhanced dataset, it became possible to construct a highly structured table that serves as the foundation for all subsequent analyses presented in this research. Furthermore, this table enabled the creation of the interactive dashboard, which supports dynamic visualization of DNS centralization and makes it possible to intuitively simulate failure scenarios or interruptions in DNS resolution, offering an agile and effective means to explore critical dependencies within the Brazilian corporate DNS landscape.

3.2.4 Dashboard Development and Visualization

The database used to build the visualization dashboard was the final consolidated dataset, which contains all enriched information and aggregated metrics. Based on this comprehensive dataset, we developed an interactive dashboard designed to facilitate the exploration and analysis of DNS resolution patterns for Brazilian domains.

Figure 3.2: DNS concentration dashboard.



Source: The Author, 2025

As seen in Figure 3.2, one of the main features of the dashboard is an interactive map that allows users to visually identify the countries where the headquarters of organizations responsible for resolving Brazilian domains are located. Users can select countries directly on the map, providing a graphical overview of the geographic distribution of DNS providers. Additionally, there is a country list interface that enables filtering by the countries where organizations resolving Brazilian domains are based. This selection can be performed either by the country of the organization's headquarters or by the organization itself.

The aggregated metrics displayed include the total revenue and the number of domains in each sector. It is important to clarify that the revenue figures refer not to the providers themselves, but rather to the sum of the revenues of the companies whose domains are resolved by the selected DNS provider. By default, the dashboard presents overall values, but when a user makes a selection, such as a specific country or provider, these metrics are dynamically updated to reflect only the selected segment. For example, if the user selects Cloudflare, the dashboard displays the total annual revenue of all companies whose domains are resolved by Cloudflare, as well as the number of such domains.

A further refinement is the temporal visualization of revenue, which allows for the analysis of revenue by day, providing the basis for assumptions about the economic impact of DNS centralization. This functionality aggregates the daily revenue of all companies whose domains are resolved by a given provider or in a selected country. There are also

breakdowns by hour and minute, enabling highly granular and dynamic analysis. Users can apply multiple filters simultaneously, obtaining a nuanced understanding of both the degree of centralization and the potential impact of disruptions in DNS services on the Brazilian market.

3.3 Methodological Limitations and Data Challenges

It is essential to acknowledge the technical and conceptual limitations that affect both the scope and precision of the analyses conducted. Recognizing these limitations does not undermine the research; on the contrary, it enhances transparency and supports more cautious interpretations of the results.

One of the first constraints observed lies in the reliance on public search engines to identify the official domains of companies. Although the use of customized queries in `duckduckgo-search` allowed for the retrieval of institutional domains with a reasonable success rate, there are cases in which companies operate multiple domains, use generic domains for local operations, or adopt subdomains for specific purposes. This can lead to the identification of non-representative domains or, in some cases, to unsuccessful automatic matches. Furthermore, validating domains through HTTPS requests does not guarantee full website functionality or its institutional representativeness. Some companies may have temporarily redirected sites, employ firewall protections, or enforce security policies that block automated script access.

Despite the adoption of artificial intelligence for sector mapping, it is important to highlight certain limitations. Some European companies could not be confidently classified into any of the established sectors and were therefore assigned to the category "Other." Additionally, within the one thousand largest companies identified in the EU, none were categorized under the Education sector, a marked difference compared to the Brazilian dataset, where educational institutions were present among the top-ranked companies.

Regarding the collection of DNS records, there are also inherent limitations in using public resolvers and local libraries such as `dns.resolver` and `ipwhois`. These tools are subject to response instability, rate limits, and potential inconsistencies in NS and A records.

The RDAP data enrichment stage, while technically sophisticated, presents semantic challenges. Identifying the controlling entity of an IP block is not always straight-

forward, particularly when the registration involves intermediaries such as Content Delivery Networks (CDNs), cloud providers, or registry organizations. In some cases, the organization name extracted from RDAP may refer to a technical entity and may not reflect the commercial or contractual relationship with the target company.

Finally, it is important to note the absence of continuous temporal verification. DNS and RDAP queries were conducted at specific points in time, which does not capture variations over time. Changes in DNS infrastructure are common, particularly among companies with a strong digital presence, and could alter the results if collected during a different time window.

In summary, although these limitations do not invalidate the study, they pose important barriers to the unrestricted generalization of the results. To mitigate their effects, strategies such as sample-based manual validations, fallback methods (`e.g.`, base domain trimming), and cross-referencing with multiple variables were employed. Nevertheless, it is recommended that future research complement this approach with longitudinal monitoring, multiple resolution sources, and, when possible, validation using internal corporate data.

4 RESULTS AND DISCUSSION

This chapter presents an analysis of the DNS resolution infrastructure of leading companies in Brazil and the EU. The study is based on a robust dataset built from the top Brazilian companies, according to the (Exame, 2025) index, and the top thousand companies in the EU, as per (CompaniesMarketCap.com, 2025), both ranked by annual revenue.

The selection of these companies covers a wide range of economic sectors, allowing the analysis to be conducted both at a general level and segmented by industry. This approach enabled an examination of the specific patterns and distribution of the DNS infrastructure in different economic contexts.

As evidenced in Tables 4.1 and 4.2, which present a summary of the distribution of domains and revenue among the main sectors defined for each company, the analyzed data set includes approximately one thousand domains in each region, Brazil and the EU. This number is slightly lower than expected due to the exclusion of some domains that could not be mapped during the data processing stage.

Table 4.1: Summary of domains and total revenue by sector in Brazil.

Primary Sector	*Domain Count*	*Revenue (Million R$)*
Financial Sector	205	1,906,038.7
Energy	111	1,599,770.6
Transport Logistics and Logistic Services	95	384,381.0
Health and Healthcare Services	85	451,890.4
Agribusiness	76	626,724.3
Technology and Telecommunications	71	195,326.7
Wholesale and Retail	64	481,577.9
Capital Goods and Industrial Equipment	57	267,284.4
Real Estate and Construction	54	99,666.4
Food and Beverages	49	859,565.3
Steel Mining and Metallurgy	48	534,643.9
Oil and Chemical	36	688,844.6
Sanitation and Environment	18	80,267.9
Education	12	24,985.3

Source: The Author, 2025

In Brazil, according to Table 4.1, there is a higher concentration of domains in the financial and energy sectors, while the remaining sectors are relatively well distributed, except for sanitation and environment, and education, which show reduced participation. It should be noted that the sector categorization was based on the Brazilian industry clas-

Table 4.2: Summary of Domains and Total Revenue by Primary Sector in EU

Primary Sector	*Domain Count*	*Revenue (Million USD)*
Capital Goods and Industrial Equipment	193	2,172,778.5
Technology and Telecommunications	140	892,451.3
Financial Sector	130	1,583,661.5
Real Estate and Construction	88	598,634.9
Wholesale and Retail	84	875,991.0
Health and Healthcare Services	80	477,643.5
Others	69	251,878.9
Energy	53	758,995.0
Transport Logistics and Logistic Services	45	483,279.5
Food and Beverages	43	251,981.6
Oil and Chemical	39	823,357.7
Steel Mining and Metallurgy	29	280,696.4
Sanitation and Environment	4	66,091.3
Agribusiness	3	27,560.1

Source: The Author, 2025

sification, used as a reference for the entire dataset. At the same time, data from the EU serves primarily for comparative purposes. Thus, all companies in the Brazilian dataset were correctly mapped according to the categories present in the original financial data; on the other hand, some companies in the EU could not be classified into the defined sectors.

In the EU, as shown in Table 4.2, there is a significant concentration of domains in capital goods and industrial equipment, technology and telecommunications, and finance, while sectors such as sanitation and environment, and agribusiness occupy marginal positions, with a limited number of companies. These differences reflect both the structural characteristics of each region's economy and the limitations imposed by the domain-mapping process itself, as well as the availability of sectoral information.

The analysis involved collecting and consolidating records of NSs and ASs, supplemented by additional information extracted from company domains. Furthermore, financial metrics from the analytical process described in Chapter 3 are incorporated, ensuring an integrated approach that combines technical and economic aspects.

This approach enabled a deeper understanding of the diversity, concentration, and potential risks associated with the DNS infrastructure of the analyzed companies, contributing to the discussion on resilience, technological dependence, and digital sovereignty in the Brazilian context.

4.1 Analysis of the Distribution of Resolution Infrastructure

The results presented in this section are organized into three main dimensions: `(i)` the distribution of name servers across different ASs; `(ii)` the geographical distribution of DNS providers and its implications for dependency and resilience; and `(iii)` the concentration of DNS service providers across the defined economic sectors. Each of these dimensions is analyzed from both technical and sovereignty perspectives, highlighting how infrastructure choices made by companies and sectors translate into broad risks and dependencies. Special attention is given to the interaction between global providers, local providers, and the exposure of critical sectors to potential single points of failure or foreign control.

4.1.1 Distribution of NSs across ASs

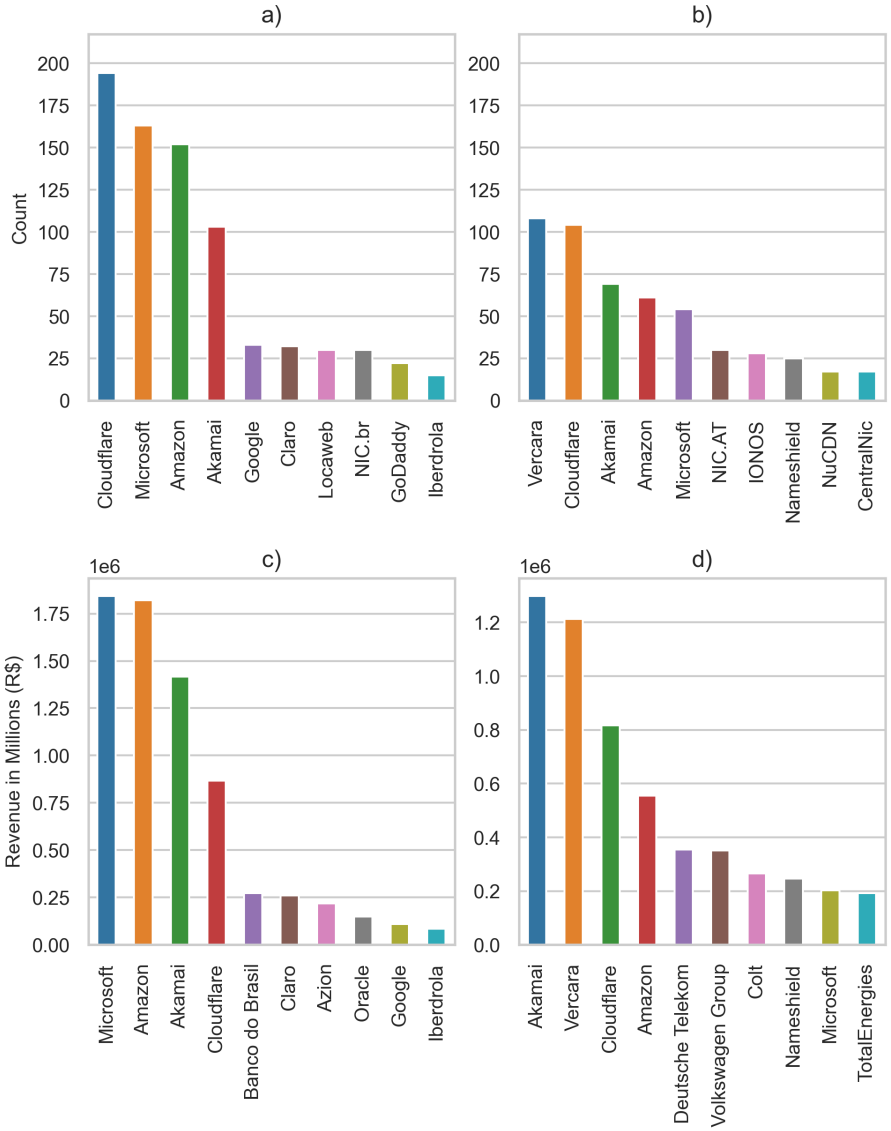
An essential aspect of DNS infrastructure resilience is the distribution of authoritative NSs across distinct ASs. A diversified AS distribution reduces the risk of single points of failure, as disruptions in a particular AS would only affect a portion of DNS resolution capacity.

The DNS infrastructure analysis was carried out using a consolidated database that aggregates information on DNS resolution and the economic data of each company analyzed. For each domain, all responsible name servers were identified, and among them, the dominant provider was determined. The dominant provider is the one responsible for the largest share of traffic or resolution for the domain, although it is important to recognize that multiple providers may simultaneously serve the same company.

Based on this mapping, it was possible to associate each domain with its primary AS, allowing the identification of the country of origin, the controlling organization, and the role of each provider as the main supplier of DNS resolution for the analyzed set. The definition of the dominant provider considered the multiplicity of suppliers, always prioritizing the one with the greatest reach in the context of the domain under study.

The analysis of the companies in this study, which includes the top 1,000 companies in Brazil and the EU by revenue, reveals a significant concentration of DNS services among a few dominant providers. Figure 4.1 shows the top providers by the number of domains they serve and the top providers by the aggregated revenue of the companies using them.

Figure 4.1: Distribution of top DNS providers in Brazil (a)(c) vs in EU (b)(d) by count and revenue.



Source: The Author, 2025

When considering the count of domains served, Cloudflare emerges as the dominant provider, with 194 domains across the studied companies. It is followed by Microsoft (163), Amazon (152), Akamai (103), and Google (33). National providers, such as Claro (32) and NIC.br (30), also appear in this distribution, though with more localized reach. Google, Claro, Locaweb, NIC.br, and GoDaddy are notable in the Brazilian market, but remain behind the global leaders. Claro, Locaweb, and NIC.br are national companies, while the top four are North American.

When ranked by the total revenue of the companies using each provider, the picture shifts somewhat. Microsoft leads with a cumulative revenue of approximately 1.84 billion BRL, followed closely by Amazon and Akamai with 1.82 billion BRL and 1.41 bil-

lion BRL, respectively. While Cloudflare leads in terms of domain count, it ranks fourth in revenue with 867.75 million BRL, indicating a broader reach among a larger number of smaller companies. Banco do Brasil appears as the fifth-largest in aggregate revenue in Brazil, reflecting the financial sector's relevance and the presence of national players among global giants. Other significant providers in revenue include Oracle, Google, and Iberdrola.

This distribution highlights two important trends: first, a strong reliance on a small group of major global providers, and second, a relatively limited diversity in AS usage, which could imply systemic risks in the case of failures or attacks targeting these major networks. While local and regional providers do contribute to the overall landscape, their market share remains comparatively small.

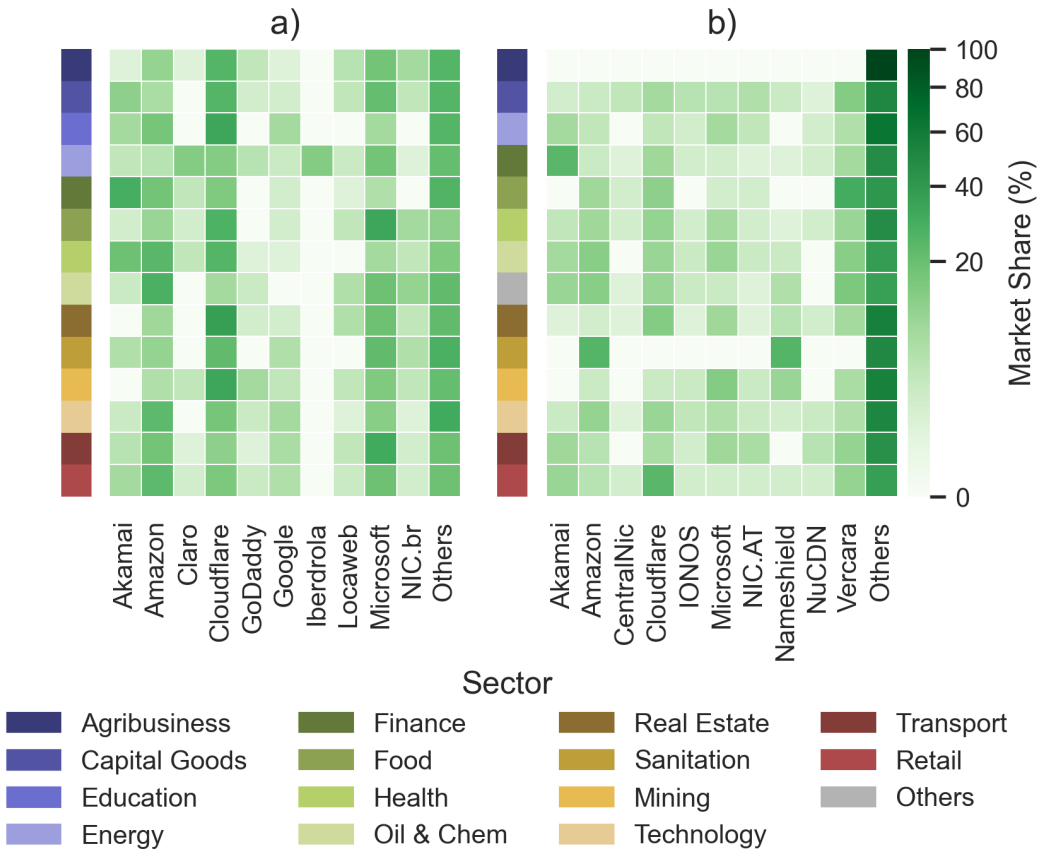
In the EU, the scenario presents considerably less concentration. Vercara and Cloudflare are among the leading providers, each with around 100 domains, followed by Akamai, Amazon, and Microsoft. From this point onward, the distribution becomes much more fragmented, with no provider surpassing 15 percent of the total number of domains, whereas in Brazil, this threshold exceeds 18 percent. This fragmentation can be observed in Figure 4.1 b).

Regarding revenue distribution, as shown in Figure 4.1 d), the EU presents less concentration, although Akamai maintains significant participation, especially in the financial sector, followed by Vercara and Cloudflare. There is a greater diversification of providers, with significant participation by companies from the European economic bloc. However, it is important to note that, despite this greater diversification, the leading providers in terms of both domains and aggregated revenue are also based in the US. Even so, European providers are able to occupy preeminent positions in the regional market, both in terms of domain count and total revenue, indicating a more balanced scenario.

4.1.2 Provider Concentration and Sectoral Dependence

The analysis of the distribution and centralization of domains by economic sector adopts two complementary approaches to reveal nuanced dynamics of DNS provider concentration among the largest companies in Brazil and the EU. The first approach examines the distribution of key providers in each sector, drawing from the providers highlighted in the previous analysis. Results are illustrated in Figure 4.2, which features heatmaps of centralization of domains in Brazil and the EU.

Figure 4.2: Dominant provider market share among sectors by count in Brazil (a) vs in EU (b).



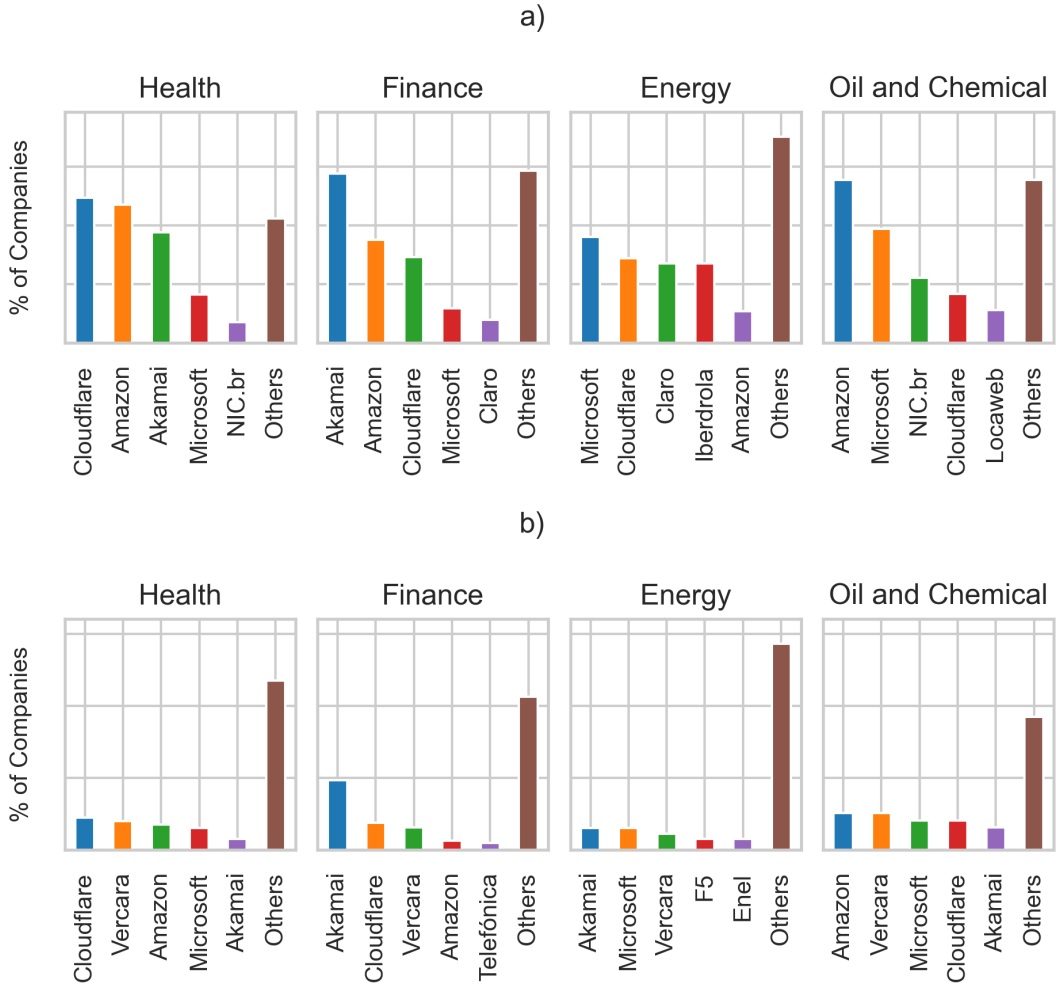
Source: The Author, 2025

A sector-level perspective reveals that while empirical analysis of these heatmaps confirms a pronounced concentration of domains in Brazil among providers such as Cloudflare, Amazon, and Microsoft, the EU is also dominated by Akamai, Amazon, Microsoft, and Cloudflare, but shows a stronger presence of regional providers. This sectoral visualization exposes trends such as the financial sector's heavy reliance on Akamai in both regions, and also makes visible the broader distribution among lower-market-share providers in Europe. Across most sectors, the dominance of four global actors is evident: Cloudflare, Microsoft, Amazon, and Akamai. Secondary providers like Google, Claro, Locaweb, GoDaddy, and NIC.br also have sectorally relevant roles.

This duality is further illustrated when analyzing both the share of domains and the share of aggregate sector revenue controlled by each provider. In Brazil, the top ten providers collectively hold a much higher share of domains than in the EU, pointing to a more concentrated provider landscape. For example, in the Financial sector, Akamai manages 29% of domains and supports 54% of the sector revenue, reflecting a large cen-

tralization on a single provider. In the Oil and Chemical sector, Amazon is responsible for % of the sector revenue and 28% of the domains, highlighting high-value concentration. Meanwhile, the Food and Beverage sector illustrates divergence, with Cloudflare serving 27% of domains but accounting for only 4% of revenue, compared to Microsoft's 16% of domains but 76% of revenue. This divergence signals a dual market structure, where smaller companies are more likely to one provider, whereas market leaders opt for another.

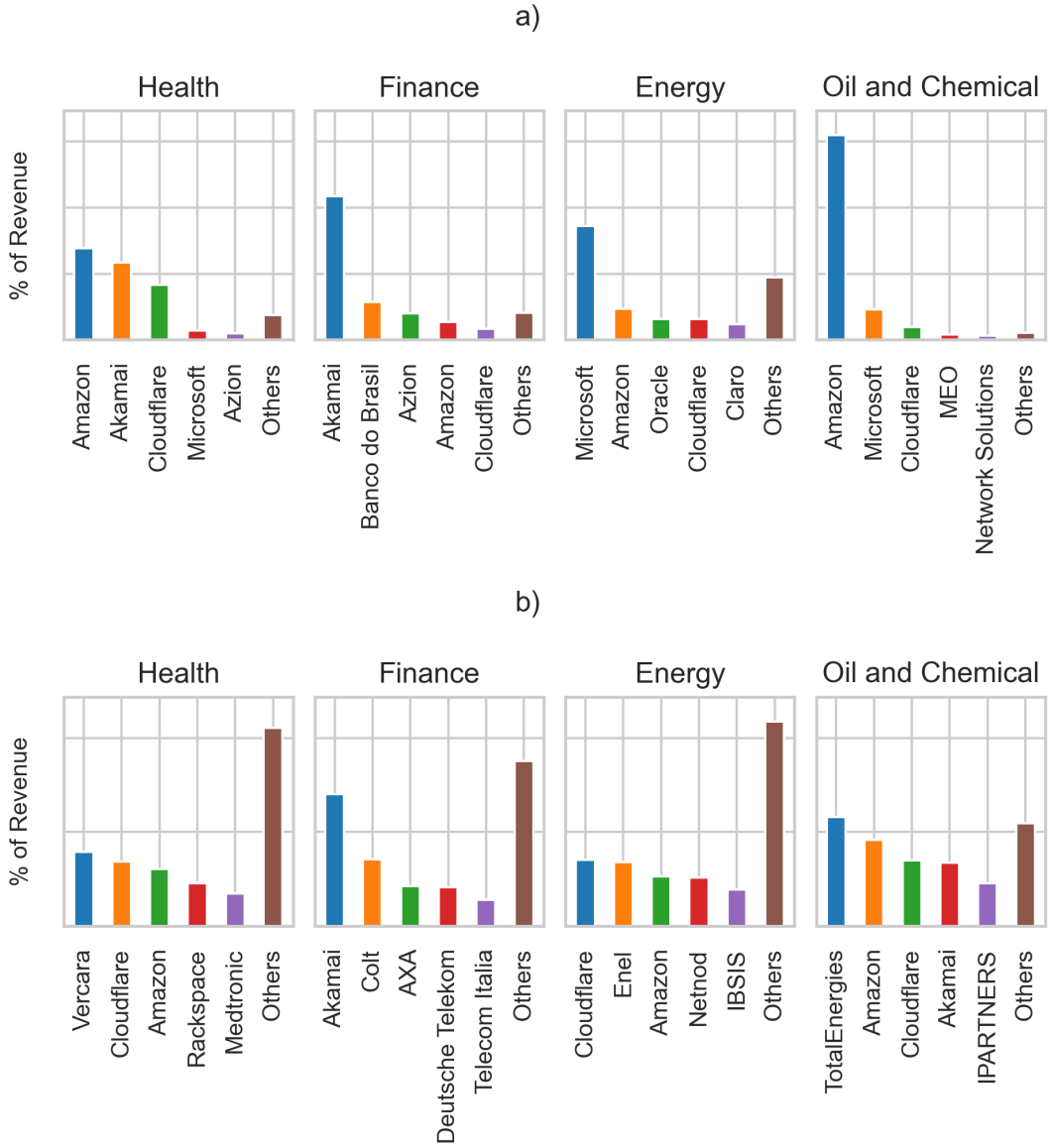
Figure 4.3: Top 5 Providers in each sector by Count in Brazil (a) vs in EU (b).



Source: The Author, 2025

Detailed sectoral analyses, as depicted in Figures 4.3 and 4.4, further reveal specialization and unique patterns. In Brazil, the health sector is divided primarily among Cloudflare, Amazon, Akamai, and Microsoft. The energy sector exhibits a more balanced provider distribution, with a notable presence of Claro and regional providers like Iberdrola, from Spain. The Oil and Chemical sector is concentrated in Amazon and Microsoft, but NIC.br and Locaweb maintain meaningful roles.

Figure 4.4: Top 5 Providers in each Sector by Revenue in Brazil (a) vs in EU (b).



Source: The Author, 2025

In the EU, although US providers remain important, the cumulative share of regional actors is considerable, evidencing greater decentralization. Notably, in the oil sector, Total Energies (France) leads, outpacing Amazon, Cloudflare, and Akamai. This is the only sector in which a regional provider surpasses global giants in DNS resolution share.

These findings have important implications for resilience and digital sovereignty. High levels of concentration in vital sectors such as Finance, Energy, and Health introduce systemic risks, in which failures or targeted attacks against a dominant provider can trigger cascading effects throughout the entire supply chain. In contrast, sectors with a more fragmented provider landscape, such as Technology or Capital Goods, demonstrate

greater fault tolerance and enhanced governance capabilities.

By analyzing the distribution and concentration of DNS providers across different sectors in Brazil, one can observe, especially over the past two decades, the emergence of the phenomenon that (FLORIDI, 2020) describes as “de facto digital corporate sovereignty’, when large corporations begin to exercise an almost sovereign decision-making power over essential infrastructures of the digital society.

However, this corporate self-sufficiency rarely results in truly competitive markets. On the contrary, it tends to consolidate hegemonic positions or even monopolies, weakening the diversity of options and, consequently, shifting the decision-making power from users and clients to the corporations themselves. This aligns with the scenario observed in Brazil: in sectors where provider concentration is higher, dependence on a few players translates not only into operational risks but also into limitations on national autonomy and digital sovereignty.

4.1.3 Provider Concentration Evaluation Using HHI

Another perspective on DNS market structure can be achieved through the HHI, a widely used metric by authorities such as the US Department of Justice, particularly by the Antitrust Division, for market analyses and merger guidelines. As presented in 2.4, the HHI ranges from 0 to 10,000, where a score of 10,000 indicates a pure monopoly and total lack of competition. According to (U.S. Department of Justice, 2024), an HHI above 1,000 is considered concentrated, with values between 1,000 and 1,800 representing moderate concentration and above 1,800 indicating a highly concentrated market.

As shown in Table 4.3, which presents the HHI index for the Brazilian DNS resolution market, the analysis considers both the market share by number of domains and by the aggregated revenue of companies whose domains are resolved by each provider. In the Brazilian market, every sector shows at least some degree of concentration. Certain sectors, such as Food and Beverages, Real Estate and Construction, and Education, are classified as highly concentrated according to their HHI values. Other sectors are moderately concentrated. The overall HHI for all companies and sectors, based on domain count, is 1,053.12, just above the threshold for being flagged as a concentrated market. When considering revenue, concentration levels are even higher, with most sectors classified as highly concentrated according to this metric.

However, it is important to note a limitation of this analysis: the calculation is

Table 4.3: DNS provider HHI by sector in Brazil.

Economic Sector	*HHI by Provider Count*	*HHI by Revenue*
Agribusiness	1,246.59	1,667.11
Capital Goods and Industrial Equipment	1,338.50	1,773.55
Education	1,805.11	2,829.40
Energy	1,013.35	2,215.50
Financial Sector	1,449.17	3,339.44
Food and Beverages	2,019.56	6,091.23
Health and Healthcare Services	1,634.88	2,513.19
Oil and Chemical	1,450.66	6,144.16
Real Estate and Construction	1,927.55	2,675.05
Sanitation and Environment	1,358.20	2,022.35
Steel Mining and Metallurgy	1,562.21	2,329.93
Technology and Telecommunications	1,093.44	1,320.62
Transport Logistics and Logistic Services	1,516.93	1,704.65
Wholesale and Retail	1,298.58	1,977.60
Total	1,053.12	1,438.18

Source: The Author, 2025

based only on the top 1,000 companies in Brazil by revenue. The dataset for each sector is therefore not exhaustive, and sectors with few representatives among the top companies, such as Sanitation and Environment, may display artificially high HHI values. Nonetheless, these figures are useful for comparison with the EU, whose results are presented in Table 4.4.

Comparatively, HHI values in the EU are substantially lower than those in Brazil. Most sectors in the EU do not approach the threshold for market concentration, with the exceptions of Agribusiness, Food and Beverages, and Sanitation and Environment. As with the Brazilian data, these cases tend to reflect sectors with fewer entities in the dataset. When looking at aggregate value, the HHI for all companies and sectors in the EU is 395.23, which indicates a highly decentralized market structure, especially when compared to Brazil.

This comparison demonstrates a greater centralization of the Brazilian DNS resolution market around a few providers. These key providers in Brazil are predominantly foreign, raising additional concerns regarding digital sovereignty, since the infrastructure is subject to external jurisdiction and governance.

A comparative perspective can be drawn by referencing the longitudinal study conducted by (BATES et al., 2021), which analyzed the historical evolution of DNS provider concentration among the Alexa Top 1,000 sites from 2011 to 2017. The find-

Table 4.4: DNS provider HHI by sector in EU.

Economic Sector	*HHI by Provider Count*	*HHI by Revenue*
Agribusiness	2,221.78	7,769.08
Capital Goods and Industrial Equipment	428.01	929.51
Energy	338.48	800.71
Financial Sector	778.96	1,221.77
Food and Beverages	1,281.90	3,758.87
Health and Healthcare Services	484.38	857.20
Oil and Chemical	729.55	1,395.16
Others	753.81	1,642.98
Real Estate and Construction	506.44	827.97
Sanitation and Environment	2,500.00	5,573.10
Steel Mining and Metallurgy	606.54	1,121.80
Technology and Telecommunications	389.38	711.87
Transport Logistics and Logistic Services	513.40	1,844.56
Wholesale and Retail	895.45	2,566.45
Total	395.23	533.58

Source: The Author, 2025

ings align with the patterns observed in the present analysis of the top 1,000 Brazilian companies, especially regarding the leading DNS providers. In 2017, the DNS market leadership was disputed between AWS and Cloudflare, with other players such as Akamai and Dyn occupying secondary positions. Both AWS and Cloudflare held approximately 17% market share at that time.

In terms of concentration as measured by the Herfindahl-Hirschman Index, Bates observed a substantial increase over the historical series. In 2011, the HHI for the top 1,000 domains was close to 100, while by 2017, it had surpassed 700. This represents an almost sevenfold increase in concentration within that period. This trend reflects the growing dominance of major DNS providers, who have significantly expanded their market share over the years. These findings reinforce the results of the present study, illustrating that DNS resolution markets exhibited a consistent tendency toward increased concentration, underlining the relevance of continuous monitoring of provider centralization for digital sovereignty and systemic resilience.

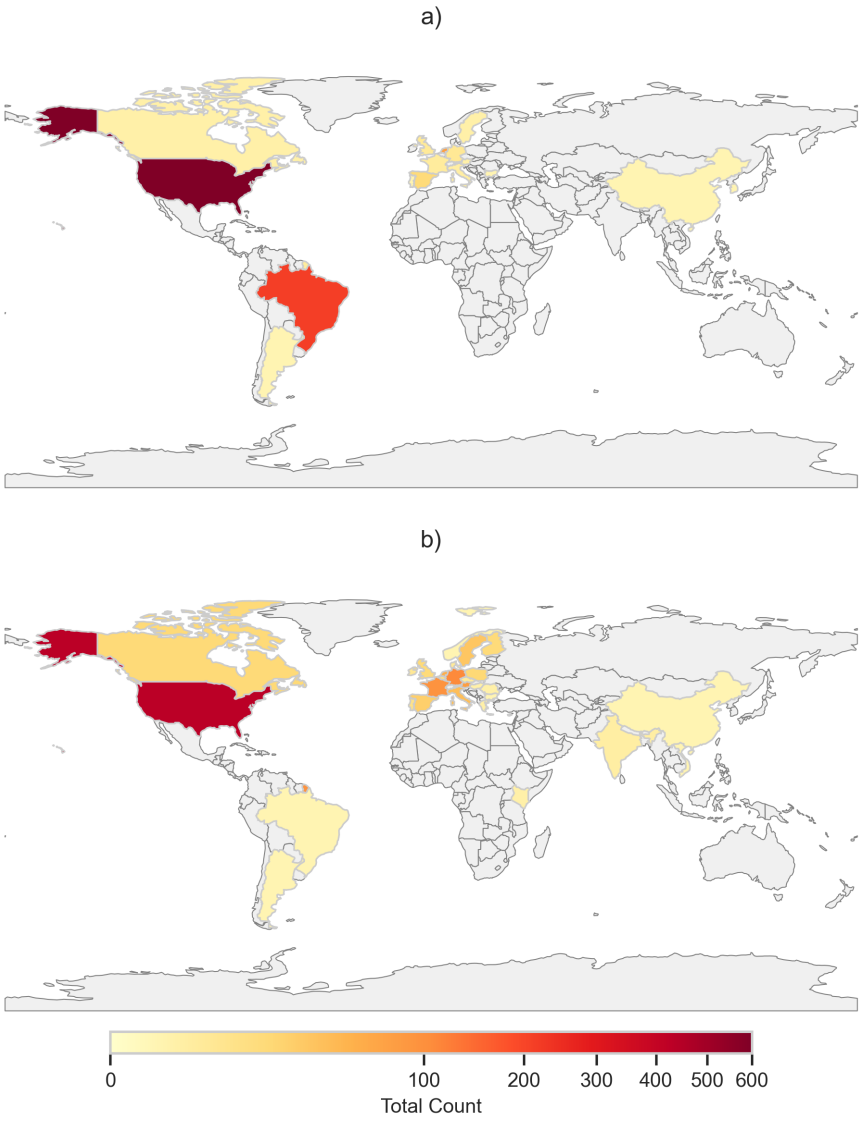
4.1.4 Geographical Distribution of Name Servers

The analysis of the geographic distribution of DNS providers was based on the centralized dataset, which consolidates all NSs associated with each domain in order to

determine the dominant provider for each company. This methodological choice reflects a necessary limitation of the research, as it is not possible to accurately enumerate all NSs and their respective roles, such as differentiating those responsible for IP resolution from those dedicated to email or auxiliary services. The selection of a single dominant provider per domain, therefore, was adopted to enable consistent comparison across the dataset.

Once the dominant provider for each company was identified, information from the corresponding AS registry was used to determine the country of origin of each provider. Cross-referencing these data with the concentration analyses from the previous subsections allows for a clear visualization of geographic centralization in DNS resolution services.

Figure 4.5: Geographical origins of dominant providers in Brazil (a) and in EU (b).

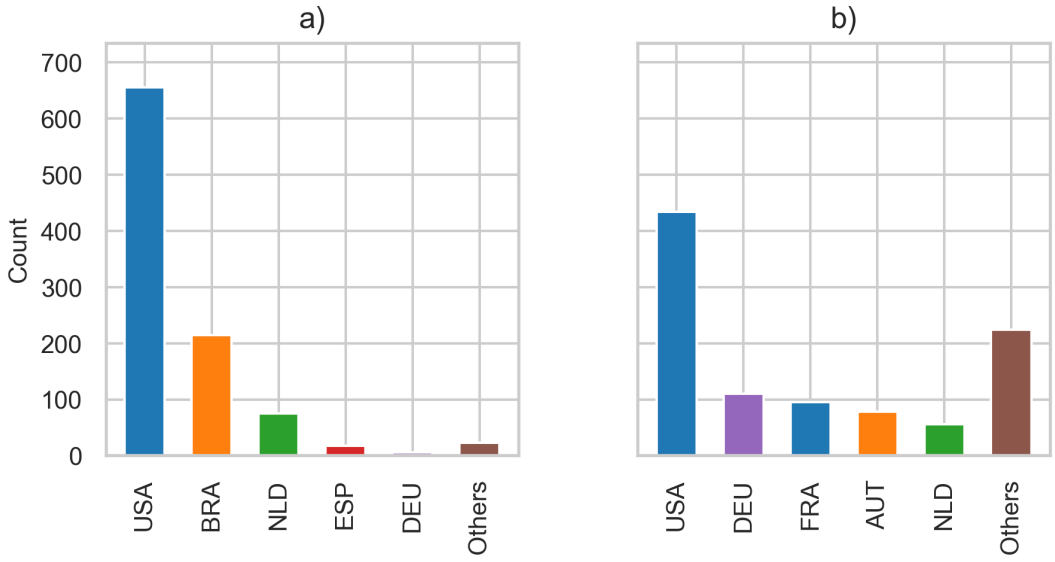


Source: The Author, 2025

As shown in Figure 4.5, which displays the global distribution of dominant providers on a world map, it becomes evident that, in the case of Brazil, although there is a relative concentration of providers based in Brazil itself, the majority of DNS resolution for the top 1,000 companies is performed by providers headquartered in the US. This significant reliance on US-based providers introduces a high degree of external dependency for Brazilian domains. The data also indicate smaller shares for providers based in Argentina, Canada, several Western European countries, and China.

In contrast, analysis of the European scenario reveals that the US remains the single country with the highest share of dominant DNS providers among leading companies. However, there is a much broader distribution of providers among the various countries that comprise the EU. Many companies in the EU make use of national providers based within their own country, which contributes to a more balanced distribution across the economic bloc. This dynamic is further illustrated in Figure 4.6, which presents a comparative percentage breakdown of DNS provider countries for both Brazil and the EU.

Figure 4.6: Top DNS Providers by Count in Brazil (a) vs EU (b).



Source: The Author, 2025

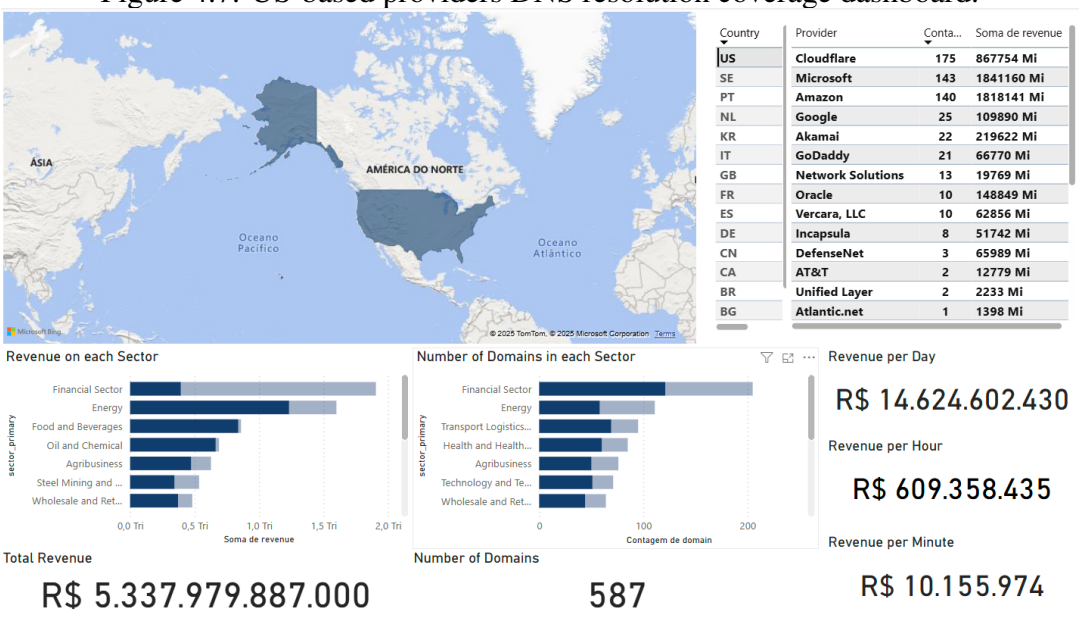
For Brazil, 66.1% of the domains belonging to the top 1,000 companies are resolved by providers headquartered in the US, followed by Brazilian companies with 21.7% and Dutch providers with 7.6%. This means that over 66% of Brazilian domains rely on US-based DNS providers, and nearly 80% are handled by foreign entities. In the case of the EU, while the US remains the country with the largest single share (43.5%), the remaining domains are much more evenly distributed among member states of the economic bloc. Notable shares are held by Germany (11%), France (9.5%), Austria (7.8%),

and the Netherlands (5.%), with other EU countries and a few additional nations accounting for smaller percentages. This comparative analysis underscores a much higher degree of geographic concentration and external dependency in the Brazilian DNS ecosystem, while the EU demonstrates a more distributed pattern of DNS resolution.

4.2 Dashboard Visualization of DNS Centralization and Economic Impact

The dashboard presented in Figure 4.7 offers a localized visualization of the share of DNS providers based in the US. This enables a clear understanding of the market share held by US-based DNS operators across multiple economic sectors, both in terms of the number of domains and the total revenue involved, as well as identifying the principal US providers. The data reinforces previous findings, confirming the US as the dominant country in the DNS resolution landscape for major Brazilian companies.

Figure 4.7: US-based providers DNS resolution coverage dashboard.



Source: The Author, 2025

To support a more nuanced vulnerability analysis, as proposed in the methodology, the study uses aggregate company revenue as a proxy for potential economic exposure. While it is important to recognize the limitations of this metric, since the relationship between DNS availability and revenue is not linear, nor does an outage necessarily result in a direct and immediate loss of revenue. No established metric currently exists to accurately quantify the impact of DNS disruptions on diverse business models and sectors. It is also acknowledged that some sectors are intrinsically more vulnerable than others. For

instance, the banking and financial sector is highly dependent on uninterrupted internet connectivity, a structural feature of the modern financial market.

Therefore, the dashboard presents revenue exposure calculations for companies whose DNS is resolved by US-based providers, broken down by day, hour, and minute. The total aggregated revenue for these companies exceeds BRL 5.3 trillion, distributed among 587 domains. This equates to approximately BRL 14.6 billion per day, BRL 609 million per hour, and BRL 10.1 million per minute. These values highlight the scale of potential economic exposure: in the event of a DNS service disruption affecting US-based providers, companies representing BRL 10.1 million in revenue would be at risk for each minute of downtime.

It is crucial to reiterate that these figures are indicative rather than predictive, given the complex and non-linear relationship between digital infrastructure availability and actual financial loss. Moreover, the revenue of companies is not distributed evenly over time; there are seasonal peaks where the impact of outages would be amplified. For example, e-commerce companies are especially vulnerable during peak sales periods such as Black Friday, Christmas, or other major shopping events. Any DNS resolution failure during these times could result in financial losses far exceeding the baseline metrics presented here.

In summary, the dashboard not only quantifies the degree of economic dependency on US-based DNS providers but also underscores the magnitude of systemic risk associated with such concentration. It highlights the necessity of contingency planning and the potential benefits of diversifying DNS resolution providers as a strategy for risk mitigation and improved digital resilience.

Using the dashboard's interactive visualizations, it was possible to simulate a scenario in which the participation of the provider Cloudflare is isolated within the Brazilian DNS resolution landscape. Cloudflare, a US-based company, stands out as the provider responsible for the largest number of domains among the top Brazilian companies, serving 175 domains in total. The aggregated revenue of the companies relying on Cloudflare's DNS services amounts to BRL 867.3 billion.

Applying the same revenue breakdown methodology as in previous analyses, Cloudflare's client portfolio corresponds to an average daily revenue exposure of BRL 2.3 billion, BRL 99 million per hour, and BRL 1.6 million per minute. This metric quantifies the scale of economic dependency on a single foreign provider for digital infrastructure continuity.

Examining sectoral distribution, Cloudflare's presence is broad, but the largest revenue concentrations are found in the wholesale and retail, agribusiness, and energy sectors. In terms of domain count, Cloudflare's highest concentration is within the financial sector, with 30 domains, followed by agribusiness (21 domains) and real estate and construction (20 domains). This distribution reveals that Cloudflare's market share is not limited to a specific segment, but rather spans a wide cross-section of the Brazilian economy.

This focused analysis demonstrates that a significant portion of the Brazilian corporate sector is dependent on a single foreign DNS provider. Any technical or geopolitical incident affecting Cloudflare could, therefore, pose substantial operational and financial risks, potentially impacting multiple critical economic sectors simultaneously.

A third scenario examined through the dashboard's dynamic visualization focuses on the provider Akamai, a US-based company with multiple international subsidiaries, including significant operations in the Netherlands. Within the Brazilian context, Akamai is the fourth largest DNS resolver by number of domains, responsible for 70 domains with a combined client revenue of BRL 1.4 trillion.

Akamai's relevance in the Brazilian DNS market stems less from its broad cross-sectoral presence and more from its overwhelming dominance in the financial sector. Of the 70 domains managed by Akamai, 59 belong to companies in the financial sector, representing 28.7% of all domains in that segment. More striking, however, is Akamai's share of sectoral revenue: the total revenue of financial sector companies using Akamai for DNS resolution reaches BRL 1 trillion, out of an overall sector revenue of BRL 1.9 trillion. This means that Akamai is responsible for resolving DNS for companies accounting for more than 50% of the entire Brazilian financial sector's revenue.

This level of concentration is particularly significant given the critical and sensitive nature of the financial sector, which is both a high-value target for cyberattacks and deeply reliant on digital infrastructure for uninterrupted service delivery. As highlighted in previous chapters, such concentration increases systemic risk, as technical failures or targeted incidents affecting Akamai could disrupt a substantial share of the national financial system's digital operations. To quantify this vulnerability, revenue breakdowns reveal that companies relying on Akamai collectively generate approximately BRL 3.8 billion per day, BRL 161 million per hour, and BRL 2.7 million per minute.

These figures illustrate the potential scale of exposure and the overwhelming reliance of the Brazilian financial sector on a single foreign provider. A critical portion of

national financial DNS infrastructure is subject to external jurisdiction, governance, and potentially foreign legal or regulatory enforcement. In practical terms, this dependence could limit Brazil's capacity to ensure autonomous control, oversight, and continuity of key digital services in the event of geopolitical tensions, international disputes, or unilateral actions taken by foreign entities. Strengthening digital sovereignty would therefore require strategies to diversify provider dependency, encourage local or regionally governed DNS alternatives, and promote resilient architectures that minimize exposure to extraterritorial risks.

5 SUMMARY, CONCLUSION AND FUTURE WORK

This study analyzed the centralization of DNS services from an economic perspective, focusing on Brazil's digital sovereignty. The main objective was to quantify the economic and financial consequences arising from the dependence on digital infrastructure controlled by a limited number of global providers, particularly in the Brazilian context. To achieve this, a comprehensive methodology was proposed and implemented, involving the collection and analysis of data from the 1,000 largest companies in Brazil by revenue, as well as from the 1,000 largest companies in the EU for comparative analysis. The research included a systematic process of extracting corporate information, enriching it with technical data on DNS and network infrastructure, and constructing a structured database for further analysis. One of the fundamental contributions was the development of an interactive dashboard designed to visualize the economic distribution of DNS dependencies and to illustrate the potential impacts of centralization.

The results revealed a significant concentration of DNS services among a small group of dominant providers, predominantly headquartered in the US. This pattern was especially pronounced in Brazil. When domains were analyzed by count, Cloudflare emerged as the leading provider, followed by Microsoft, Amazon, Akamai, and Google, while national providers like Claro and NIC.br demonstrated a more limited and localized presence. When the analysis considered the aggregate revenue of the client companies, Microsoft, Amazon, and Akamai led the rankings, highlighting their relevance among the country's largest corporations. The comparison with the EU showed less provider concentration and a greater degree of diversification, with a larger number of regional providers in leading positions.

The assessment of market concentration using the Herfindahl-Hirschman Index confirmed that the Brazilian market for DNS resolution is predominantly concentrated, with most sectors classified as moderately or highly concentrated, especially when revenue is used as the metric. The overall HHI for Brazil was 1,053.12 when considering provider count and 1,438.18 by revenue, whereas the EU presented a substantially lower HHI of 395.23 by count and 533.58 by revenue, indicating a more decentralized market structure. The study also found that the geographical dependency of Brazil is considerable: 66.1% of the domains associated with leading Brazilian companies are resolved by providers based in the US, and almost 80% by foreign entities. In contrast, in the EU, although the US still leads with 43.5%, there is a much more balanced distribution among

member states.

The economic exposure resulting from this dependency is significant. When aggregate revenue is used as a proxy, Brazilian companies that depend on US-based DNS providers account for more than BRL 5.3 trillion in total revenue. The potential exposure in the event of service interruptions reaches $ 14.6 billion per day, BRL 609 million per hour, and BRL 10.1 million per minute. Case studies further illustrate these risks, such as the average daily exposure of BRL 2.3 billion associated with Cloudflare, and the situation of Akamai, which is responsible for resolving more than 50% of the revenue in the Brazilian financial sector and represents a potential exposure of BRL 3.8 billion per day.

In summary, this research highlights the urgent need for policy, technological, and operational strategies to address the risks of DNS centralization and to build greater autonomy and resilience within the Brazilian digital ecosystem. This high level of dependency on a few foreign providers raises serious concerns about digital sovereignty, resilience, and national autonomy. High concentration in vital sectors such as finance, energy, and health introduces systemic risks, where failures or attacks on a dominant provider could result in cascading effects throughout entire value chains.

Several opportunities for future research are identified. The implementation of longitudinal monitoring could allow for the capture of temporal variations in DNS infrastructure, while the use of multiple DNS resolution sources would help mitigate the instabilities and inconsistencies observed with public resolvers and local libraries. Further validation of data through internal corporate information, where available, would complement public sources and improve the accuracy of organizational attributions and the roles of name servers. The development of methodologies that better quantify the financial impact of DNS interruptions is another promising avenue, as it is necessary to account for the non-linear relationship between service availability and revenue loss, as well as sector-specific dynamics.

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